How-To Library

Informatica

Deploy the Informatica® Enterprise Data Catalog 10.5.1.1 Solution on the Google Cloud Marketplace

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Abstract

The automated marketplace solution uses Google Cloud Platform templates to launch, configure, and run the Google Cloud compute, network, storage, and other services required to deploy a specific workload on Google Cloud. This deployment reference provides step-by-step instructions for deploying Informatica Enterprise Data Catalog on Google Cloud Platform from Google Marketplace.

Supported Versions

- Enterprise Data Catalog 10.5.1.1

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Overview

Customers of Google Cloud and Informatica can execute the Enterprise Data Catalog deployment from the Google Cloud Marketplace to create an Informatica domain in the Google Cloud and explore Enterprise Data Catalog functionality.

This deployment article provides step-by-step instructions for deploying Enterprise Data Catalog on the Google Cloud. Automated reference deployments use Google Cloud templates to launch, configure, and run the Google Cloud compute, network, storage, and other services required to deploy a specific workload on Google Cloud.

Intended Audience

This article is for users who are responsible for deploying the Enterprise Data Catalog 10.5.1.1 solution on the Google Cloud marketplace.

As a user with administrator privileges to deploy applications on Google Cloud, you should be familiar with Google Cloud resources such as the Google deployment manager, network on the Google cloud, compute engine, Google storage, firewall rules, Cloud NAT, and roles. See the Google Cloud documentation.

Familiarity with the concepts of IP CIDR, public and private IP addresses is recommended.

You must also be familiar with Enterprise Data Catalog. To find Enterprise Data Catalog documentation, see the Informatica documentation portal.

Costs and Licenses

You are responsible for the cost of the Google Cloud services while running this deployment. There is no additional cost to use this marketplace deployment.

The Google deployment manager template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, affect the cost of deployment. See the pricing pages for each Google Cloud service that you plan to use for cost estimates.

This deployment requires a license for Informatica Enterprise Data Catalog. To sign up for a license, contact your organization's Informatica sales contact or Informatica Global Customer Support.

The following table lists the instance types that you can choose based on sizing requirements:

Virtual Machine	Instance Type
Database	n1-standard-4
Informatica domain	n1-standard-8
Bastion server	n1-standard-4
Informatica cluster	n1-standard-8

Apache Log4j RCE Vulnerability

Informatica 10.5.1.1 addresses the Log4j RCE vulnerabilities by removing the JndiLookup class from the vulnerable Log4j libraries.

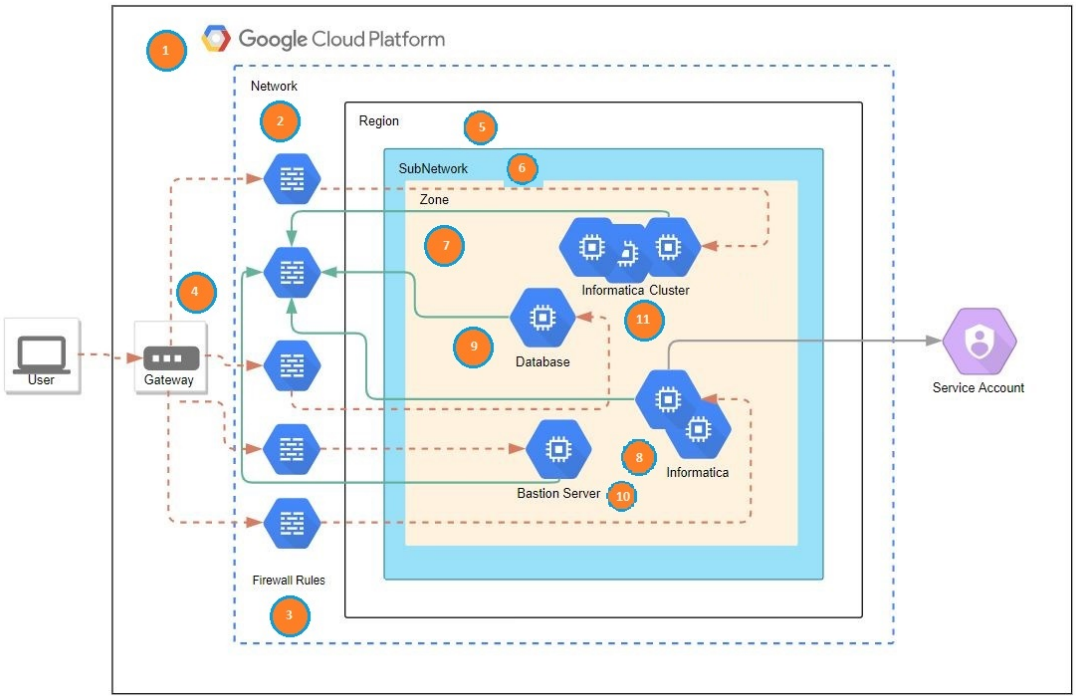
For information about the vulnerabilities, see Informatica Response to Apache Log4j RCE Zero Day Vulnerability.

Architecture

The Google Cloud Platform (GCP) marketplace solution, when you deploy it on a network, creates and connects the following resources in the network:

- Informatica domain server on a GCP Compute Engine with one additional GCP Compute Engine for High Availability.
- Informatica clients on a bastion server (optional).
- Microsoft SQL Server 2017 on Microsoft Windows Server 2016 for Informatica domain repositories. Licensing cost charged by Google.

The following image shows the architecture of the Enterprise Data Catalog solution on the Google marketplace:



The following list describes the components in the architecture:

1. The Google Cloud Platform (GCP).
2. A network to contain the Enterprise Data Catalog deployment.
3. Firewall rules to control access.
4. A gateway to control external access.
5. Region in which the solution is deployed.
6. Subnets to contain specific elements of the deployment.
7. Zone to contain database, Informatica domain, and bastion server.
8. The Informatica domain to contain Informatica services.
9. Relational database to contain the following Informatica repositories:
   - Domain configuration repository
   - Model repository
10. Bastion server, if you choose to configure one, to serve as an access point for clients.
11. Informatica Cluster.

Google Cloud Platform Resources in the Deployment

The deployment process creates or includes the following components:

- Compute engines for the Informatica domain.
- Licenses Windows 2016 server running Microsoft SQL Server 2017.
- A compute engine bastion server.
- Compute engines for the Informatica cluster.

Components in a Deployment on an Existing Network

The following components are created when you deploy the software:

Component	Description
Firewall rules	The following firewall rules are created: - One rule for the relational database server. - One rule for the Informatica domain instance. - One rule to support remote login to the bastion server. - One rule for internal communication. - One rule for the Informatica cluster.
Microsoft SQL Server 2017	One Microsoft SQL Server 2017 instance on Microsoft Windows Server 2016 running on n1-standard-4.
Virtual Machine	- One GCP Compute Engine running on n1-standard-8 for Informatica Server. - One GCP Compute Engine running on n1-standard-8 for Informatica services on High Availability (optional). - One GCP Compute Engine running on n1-standard-4 for bastion server. Bastion server is optional. - Up to six GCP Compute Engines running on n1-standard-8 for Informatica Cluster nodes.

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where the solution deploys. It can also act as a remote Windows server which you can log in to run Informatica clients.

Informatica Domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

Informatica domain includes the following application service:

Model Repository Service

The Model Repository Service manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

Content Management Service

The Content Management Service is a component within the Informatica domain that manages reference data and is responsible for compiling rule specifications into mapplets. The Content Management Service

provides reference data information to the Data Integration Service and to the Developer tool and Analyst tool. The Content Management Service stores reference data in a database that you specify.

Analyst Service

The Analyst Service is a component within the Informatica domain that runs the Analyst tool in the Informatica domain. The Analyst Service manages the connection between the service components and the users who log in to Analyst tool. You can perform column and rule profiling, manage scorecards, and manage bad records and duplicate records in the Analyst tool. The Analyst Service stores profiling, scorecarding, and bad and duplicate record data in databases that you specify.

Informatica Cluster Service

The Informatica Cluster Service runs and manages the Catalog Service and all the services associated with it. The associated services include Nomad, MongoDB, Solr, and PostgreSQL.

Catalog Service

The Catalog Service runs Enterprise Data Catalog and manages the connections between service components and external applications.

Domain Repositories

Informatica repositories, created on Microsoft SQL Server databases, store metadata about domain objects. Informatica domain includes the following repositories:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings, profiles, mapping specs, rules, and workflows.

For more information, see the Informatica Application Service Guide.

Informatica Clients

You can use several different clients with Enterprise Data Catalog:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Analyst tool

The Analyst tool is a web-based application that enables users to view and update records and clusters in a human task.

Bastion server

You can optionally deploy a bastion server as an element in the solution. The bastion server is a Google Cloud Platform (GCP) Compute Engine Windows instance installed with the Developer tool and the command line interface clients. The bastion server acts as a firewall for access to the network. It can also act as a remote Windows server that you can log in to run Informatica clients.

Catalog Administrator

The Enterprise Data Catalog administration tool that you can use to manage and monitor the resources, schedules, attributes, and connections.

Enterprise Data Catalog

The tool that you can use to perform the following tasks:

- Search for assets in the catalog.
- Verify the quality of data such as profiling information.
- View the lineage and relationship of assets in the catalog.
- Enrich assets with attributes for ease of search.

Before You Begin

Before you launch the automated deployment on the GCP Marketplace, verify the prerequisites and make the choices described in this section.

License Key Prerequisite

Verify that you have a license to deploy Enterprise Data Catalog.

You supply the license key value in the Informatica Enterprise Data Catalog License Key parameter when you configure the deployment.

Network Prerequisites

Before you deploy Enterprise Data Catalog in the Google Cloud, verify that you have completed the network prerequisites for the Informatica application services.

The automated deployment builds a new Google Cloud environment consisting of the firewall rules and other infrastructure components, and then deploys Enterprise Data Catalog into this environment.

Observe the following prerequisites when you choose to deploy into an existing network:

- If you don't intend to use public IP, verify that the Google private access is enabled.
- Verify that the following components are created:
  - Cloud NAT gateway for the deployment zone.
  - If public IP association is not required, select a subnetwork that is in the same zone that includes the Cloud NAT.
- Ensure that the selected network has a Cloud NAT on the selected zone while deploying the product without a public IP address.
- The deployment uses the existing network and subnetwork. Ensure that the selected subnetwork is enabled for Google private access while deploying the product without a public IP address.

To create a new Google Cloud network and subnetwork, follow the steps:

1. Log in to Google Cloud console.

2. In the side bar menu, navigate to **VPC network** > **VPC networks** > **CREATE VPC NETWORK**.
3. Enter the details.
4. Click **Create**.

To create a Cloud NAT, follow the steps:

1. Log in to Google Cloud console.
2. In the side bar menu, navigate to **Network Services** > **Cloud NAT** > **Get Started**.
3. Enter the details.
4. Click **Create**.

Account and Security Prerequisites

Observe the prerequisites in this section to verify that you established the required Google Cloud account and the requisite account private keys.

Google Cloud Account and Region

If you do not have a Google Cloud account, create one.

Use the region selector to choose the GCP Compute Engine region where you want to deploy Enterprise Data Catalog.

If necessary, request a service level increase for the GCP Compute Engine instance type you plan to use for the Informatica domain. You might need to increase the service level if you already have an existing deployment that uses this instance type, and you might exceed the default limit with this reference deployment.

GCP Compute Engine APIs

Verify that the following APIs are enabled:

- Compute Engine API
- Cloud Deployment Manager V2 API
- Cloud Runtime Configuration API
- Cloud Dataproc API
- Identity and Access Management (IAM) API
- Cloud Resource Manager API
- Google Cloud DNS API
- Service Usage API
- Cloud Storage
- Google Cloud APIs
- Google Cloud Storage JSON API
- IAM Service Account Credentials API

GCP Compute Engine Roles

Verify that the GCP Compute Engine project associated with your deployment has the required roles enabled. You can identify the project with the <project-number>@cloudservices.gserviceaccount.com service account.

Choose to enable one of the following roles for the project:

- Owner role

- The following set of minimum required roles:
  - Deployment Manager Editor
  - Deployment Manager Type Editor
  - Project IAM Admin
  - Service Account Admin
  - Service Account Key Admin
  - Compute Admin
  - Cloud RuntimeConfig Admin
  - Dataproc Administrator
  - Dataproc Worker
  - Compute Storage Admin
  - Storage Admin

Note: Because the <project-number>@cloudservices.gserviceaccount.com service account might be used by the deployment manager for other projects in your account, it requires specific roles.

Deploying the Enterprise Data Catalog solution on the Google Cloud Marketplace

The automated deployment of Enterprise Data Catalog on the Google Cloud Marketplace uses the GCP Compute Engine template to guide your choices and launch the solution deployment.

Use the GCP Compute Engine console to specify the deployment name and to enter parameter values to the template that you use to deploy the application. A deployment is a logical representation of a GCP Compute Engine deployment.

Specify the identifiers for the network, for the subnetwork that contains the GCP Compute Engine instances where the deployment process deploys the application components, and for the subnetwork where the deployment process creates the Model repository. The deployment process provisions the application components and creates the Model repository based on the values that you specify.

Zone and subnetwork

The automated Google marketplace solution deploys the components and services for Enterprise Data Catalog into a GCP Compute Engine network configured in the selected zone.

In the first subnetwork, the deployment process provisions the following Enterprise Data Catalog domain components on a Google Compute Engine within the network:

- Data Integration Service
- Model Repository Service
- Content Management Service
- Analyst Service
- Catalog Service
- Informatica Cluster Service

For more information about domain components, see the Informatica Administrator Guide and the Informatica Application Service Guide.

In the second subnetwork, the solution deploys the Microsoft SQL Server database for the Model repository. The Model repository stores metadata that the Informatica services create.

Step 1. Get a License

This deployment requires a license for Informatica Enterprise Data Catalog.

To sign up for a license, contact Informatica. You need the key file at the time of deployment as an input value.

Step 2. Launch the Deployment

You are responsible for the cost of the GCP Compute Engine services used while running this deployment. There is no additional cost for using this deployment. See the pricing pages for each GCP Compute Engine service that you plan to use for full details. Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Log in to the Google Cloud marketplace using the Google Cloud account.
2. Search for **Informatica Enterprise Data Catalog 10.5.1.1**, and select the link.

   The **Overview** page appears.
3. Click **Launch**.

Step 3. Specify Deployment Details

When you choose to deploy the Enterprise Data Catalog solution to an existing network, you supply the following parameters on the **Specify Deployment Details** page.

Deployment Name

Enter a name for the deployment. The name can include letters (a-z), numbers (0-9), and dashes (-).

Networking

Enter values for the following parameters:

Parameter	Description
Zone	Required. Zone in which the solution is deployed.
Network	Network that contains the deployment.
Subnetwork	The IP address for the subnetwork. **Note:** This subnetwork must be in the network identified in the Network parameter.
CIDR IP Range	Required. The CIDR IP range that is permitted to access the Informatica domain and the Informatica cluster. We recommend that you use a constrained CIDR range to reduce the potential of inbound attacks from unknown IP addresses. For example, to specify the range of 100.20.30.20 to 100.20.30.23, enter 100.20.30.20/30.
Deploy Bastion Server	Check to deploy a bastion server to access other resources in the network. Default is false.
Enable Public IP	Check to deploy compute resources with public IP addresses. Default is true.

Informatica Enterprise Data Catalog Details

Enter values for the following parameters:

Parameter	Description
Deployment Type	Size of the Informatica cluster that you want to create. Default is Small.
Informatica Enterprise Data Catalog License Key	Required. Enterprise Data Catalog License key value.
High Availability Deployment	Indicates whether you want to enable high availability for the Enterprise Data Catalog deployment. Default is false. For information about high availability for the Informatica domain, see the Informatica documentation

Confirm necessary GCP APIs are already enabled

After you configure the parameters, confirm that the necessary GCP APIs are enabled, and then click **Deploy** to launch the deployment.

Verify Solution Deployment

After click **Deploy**, the deployment detail page with current deployment status of each component and output section appears.

Output Properties

On the **Output** tab, you can view the list of resources that the deployment process creates in the network. The following table lists the outputs:

Output	Description
Informatica Administrator URL	URL:port to access the Administrator tool.
Informatica LDM Admin URL	URL:port to access the Catalog Administrator.
Informatica LDM Catalog URL	URL:port to access Enterprise Data Catalog.
Informatica HA Node IP	IPv4 address of the high availability or failover node for the domain, if you chose to enable high availability.
Informatica Administrator Tool Username	User ID of the Administrator tool client user.
Informatica Administrator Tool Password	Password for the Administrator tool user.
Informatica Database Users	User IDs for the database users of the domain databases created in the deployment.

Output	Description
Informatica Database Password	Password for the database users of the domain databases created in the deployment.
Informatica Bastion Server IP	The IP address of the Bastion server if you chose to deploy a Bastion server.

Use the URL, user ID, and password to access the Administrator tool to perform administrator tasks. For a description of the administrator tasks, see the Informatica Administrator Guide and the Data Engineering Administrator Guide.

Logs

After the completion of Enterprise Data Catalog deployment, consult logs to see the success or failure of solution element creation.

You can access the command execution log that records the following events on the Informatica domain VM:

- Creation of Informatica services.
- Data Integration Service recycling to register all changes.

If any of the tasks fail to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

For Informatica domain and service configuration related logs, check the following files

```
/opt/Infa_OneClick_Solution.log
```

For deployment related logs, check the following file:

```
/opt/startup.log
```

If the files are not present, go to the **VM Instances** page and navigate to **Informatica domain VM** > **Serial port 1 or Serial port 0 (console)** for the VM initialization logs that Google creates for the detailed diagnostics.

Author

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